

MUHAMMAD DANIYAL

Data Engineering · Real-Time Pipelines · Kafka · Spark · Airflow · AWS

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🐙 [GitHub](#) | 💼 [LinkedIn](#) | 🌐 [Portfolio](#)

PROFILE

Computer Science undergraduate (Class of 2027) specializing in Data Engineering, with hands-on experience building production-style real-time streaming pipelines, cloud data lakes, and automated analytics workflows end-to-end. Proficient in Python, SQL, Apache Kafka, Apache Spark, Apache Airflow, and dbt; also worked with Google BigQuery while completing Google Cloud's Data Engineering certifications (Data Lakes & Warehouses, Batch & Streaming Data Pipelines) on Coursera. Looking for a Data Engineering internship or junior role to contribute to real-world data infrastructure and learn from senior engineers.

CORE SKILLS

Languages	Python, SQL, Java, C++, Bash
Data Engineering	ETL Pipelines, Data Warehousing, Data Lakes, Aggregation, Dimensional Modeling
Streaming & Orchestration	Apache Kafka, Apache Spark (Structured Streaming), Apache Airflow, Docker Compose
Databases	MySQL, SQL Joins, Window Functions, Query Optimization
Cloud & Analytics	AWS S3, Amazon Athena, Google BigQuery, Databricks, Snowflake
Tools	Git, Docker, dbt, VS Code

EXPERIENCE

Data Science & Analytics Intern

Feb 2026 – Mar 2026

DevelopersHub Corporation (Remote)

- Built and evaluated ML classification models on real-world datasets, owning the pipeline end-to-end from preprocessing to model export with Pickle.
- Performed feature engineering and data preprocessing to improve input quality and downstream model performance.
- Collaborated inside a production-style analytics workflow, gaining hands-on experience with data processing and reporting at scale.

PROJECTS

Real-Time Stock Market Data Pipeline

[GitHub ↗](#)

Python · Kafka · Spark · AWS S3 · BigQuery · dbt · Airflow

- Built a real-time streaming pipeline ingesting live stock market data from a public REST API, publishing to Kafka and processing with Spark Structured Streaming.
- Implemented streaming transformations (price range, windowed rolling averages) and staged output to AWS S3 for durable, cost-efficient storage.
- Loaded streaming data into Google BigQuery and built a two-layer dbt model (staging and mart) producing clean, analytics-ready daily summary tables per stock symbol.
- Designed an Airflow DAG to orchestrate the batch portion of the pipeline (S3 upload, BigQuery load, dbt run) for automated, scheduled execution.
- Owned environment setup end-to-end — Dockerized Kafka broker, Windows Spark/Hadoop config, S3 and BigQuery credentials — for a fully reproducible one-script launcher.

Real-Time Cyber Threat Detection Pipeline

[GitHub ↗](#)

Python · Kafka · Airflow · MySQL · AWS S3 · Athena · Flask · Docker

- Architected a streaming pipeline using Kafka producers/consumers to ingest and process simulated web-server logs containing malicious traffic.
- Implemented a rule-based detection engine flagging SQL injection, XSS, brute-force, and directory-traversal attacks with live alerts.
- Stored structured logs in MySQL and raw JSON in S3 (partitioned) for cost-efficient querying via Amazon Athena; built a Flask dashboard with live alerts, attack statistics, and time-series visualizations.
- Automated hourly aggregation and monitoring with Airflow DAGs, with the entire stack containerized via Docker Compose for reproducible deployment.

Pakistan Flood Risk Monitoring System

[GitHub ↗](#)

Python · Apache Airflow · MySQL · Streamlit · OpenWeatherMap · FAO API

- Built an end-to-end automated ETL pipeline with Airflow, ingesting real-time weather data from OpenWeatherMap and historical flood records from FAO Pakistan.
- Designed a multi-factor risk scoring algorithm (temperature, humidity, rainfall) with automated email alerts for high-risk cities.
- Delivered a Streamlit dashboard with filters, charts, and CSV export for operational flood-risk decision-making.

Customer Transactions Data Warehouse

[GitHub ↗](#)

SQL · MySQL · Window Functions · Star Schema

- Designed a star-schema data warehouse for customer transactions, leveraging SQL window functions (RANK, LAG, running totals) for analytics.
- Optimized reporting queries with indexing and aggregations, demonstrating data modeling and query performance tuning.

EDUCATION

BS Computer Science

2023 – 2027

National Skills University, Islamabad

FSc Pre-Engineering

Completed

Punjab College